—— THE ROYAL TRIALS HAVE BEGUN

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Kibre Gugs: The Battle for the Throne



A real-time multiplayer action game inspired by Ethiopian history and art. Train as a warrior, unite under four banners, challenge the enchanted King and claim the throne.

CLICK HERE TO VISIT THE GAME

https://ethiopian-history-based-game-for-hackaton.onrender.com/

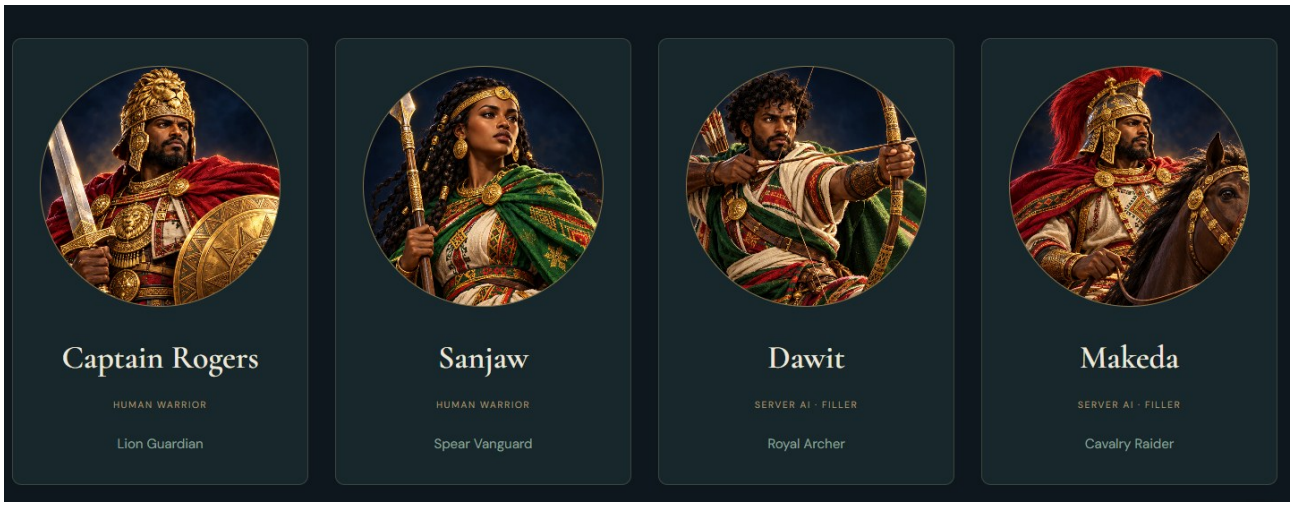
REAL-TIME COMBAT · FOUR-PLAYER SQUADS · VERIFIED GLORY · ON-CHAIN ACHIEVEMENTS

What the game is about

Kibre Gugs (“the glory of the Gugs”, the traditional Ethiopian mounted tournament) is set in **Aksumara**, a fictional kingdom drawn from Ethiopian history, landscapes and art: obelisks, highland fortresses, river fords and citadels. The kingdom has fallen under an enchanted King. Players rise from trainee to champion by fighting live battles against other players and AI, and by following a 15-mission story campaign across three chapters that ends in the Eclipse Throne finale.

Every finished match produces a tamper-evident result record (match, winner, timestamp and a Keccak-256 hash) that players can verify themselves and, when configured, attest on a public blockchain. A leaderboard, badges, tournaments and a real-world lunch perk reward skill and teamwork.

Heroes



A squad lobby: two human warriors and two server-AI fillers, one for each hero role.

Hero role	Fighting style
Lion Guardian	Sword and shield. Strong strikes, a raised-shield block and a healing rally that protects nearby allies.
Spear Vanguard	Long reach and fast thrusts that pierce a narrow line.
Royal Archer	Ranged arrows aimed with the mouse, for fighting from a distance.
Cavalry Raider	Mounted fighter with a charge ability, built for riding and jousting.

Features

Game modes

Mode	What happens
Training	A private tutorial: movement, striking, blocking and dodging.
1v1 Duel	A live duel against another player; an AI rival fills in if nobody joins.
Banner Conquest	Two four-warrior squads capture and hold banner circles.
Royal Quest	A cooperative four-player boss quest with puzzles.
Story campaign	15 missions in 3 chapters: protecting a standard-bearer, horse riding, landmarks, the wind-swept spear joust, Hold the Ford against monster waves, banner conquest, escorts, rescues, a stolen standard, night beacons, artifact and mirror puzzles, a guardian boss and the Eclipse Throne finale.
Tournament	A four-warrior knockout bracket. Empty seats are filled by AI champions; the winner earns points and the Tournament Champion badge.

Gameplay

- An authoritative server simulates every battle 30 times per second; the browser predicts your own movement so controls feel instant.
- Sword arcs, spear thrusts, arrows, a raised-shield block, dodges with brief invulnerability and a special ability for each hero.
- **Aimed spear throws:** in the joust and throughout Chapter II, aim with the cursor along a guide line; the spear leaves the hand and flies, bent by the wind.
- **Ally and enemy markers:** from Banner Conquest on, green signs float over squadmates and red signs over every enemy.
- **Hold the Ford:** defenders take reduced damage and get a “second wind” once per wave.
- Gradual difficulty: early missions send monsters one at a time with weaker blows, rising to full strength by the finale. Restart and Next Stage buttons; progress is saved after every mission.
- Physics-driven animation: running gait, wind-up and follow-through swings, spring-damped limbs, cape and weapon, and a 2.5D isometric battlefield.

Multiplayer and social

- Live matchmaking: guests play guests; identity-verified players are matched within their age band.
- Squads with invite codes and QR passes, a ready-up lobby and a team chat that opens as a chat bubble and hides during battle.
- Friends by friend code, online presence and squad invitations.
- Live notifications: a bell, an activity feed and pop-up toasts.
- Spectator mode: watch any running battle and follow any fighter.

Identity, rewards and Web3

- **Verify identity:** scan or upload a Fayda national ID and a selfie; on-device text recognition suggests the birth date and a human reviewer approves it.

- **Royal Lunch Perk** (prototype): a four-warrior squad that meets for lunch gets 40% off Chanoly noodles and the gilded Royal Feast armor. Verified accounts only.
- **Match proofs:** each result is hashed with Keccak-256 and can be re-checked in the browser from the Battle Chronicles.
- **Wallet and on-chain profile:** link a wallet by signing a one-time challenge and register a profile with an EIP-712 signature; no custodial keys.
- **Achievement tokens:** server-validated badges (First Blood, Sword Master, Thronebreaker, Tournament Champion and more) claimable as ERC-1155 tokens on Base Sepolia, with a Hall of Legends leaderboard.
- **Security:** revocable sessions, room admission guards, rate-limited chat and private, auto-deleted ID media.
- **Soundtrack:** an original generated score with kebero drums, a horn drone, a choir pad and a krar-style melody in the Tizita and Anchihoye modes.

Tech stack

Layer	Technologies
Game client	React 19, Phaser 3.90 (2.5D isometric renderer), TypeScript, Vite, Zustand, Tailwind CSS
Real-time networking	Colyseus 0.16 rooms over WebSockets (battles, campaign, squad lobbies); Socket.IO for team chat, notifications and spectators on the same port
Game server	Node.js 22, Express, zod validation and a shared deterministic simulation package used by both server and browser
Database and auth	Supabase: PostgreSQL (private game_runtime schema), email sign-in, private storage buckets and a scheduled Edge Function for media cleanup
Blockchain	Solidity and OpenZeppelin (RoyalRecords attestations, EIP-712 profile registration, ERC-1155 badges), viem, Base Sepolia, Keccak-256 hashing
On-device AI	MediaPipe Face Landmarker and Tesseract.js (ID birth-date OCR), running locally in the browser
Audio	Web Audio API: a generated soundtrack and effects, with no audio files
Quality	Vitest unit and integration tests, Playwright browser tests that play every mission, ESLint and Prettier
Hosting	One Render Docker service serving the web app, API and WebSockets; Supabase for data, auth and storage; pnpm monorepo

How it fits together

The browser loads the React app from Render and opens a WebSocket to the same service. Player inputs go to a Colyseus room, which runs the shared simulation at 30 Hz and streams snapshots back. When a match ends the server saves the result, its hash, progress and badges to Supabase in one transaction, queues the optional blockchain attestation and sends a live notification.

The team and their roles

Kibre Gugs was designed and built by a five-person team for the hackathon.

#	Team member	Role
1	Zelalem Berhanu zemberhanu@gmail.com	Research Data Scientist for game story and QA Engineer
2	Eden Abeselom edenabeselom23@gmail.com	Storytelling Specialist and Research Analyst
3	Befiker Gezahegn befikerghailemichael@gmail.com	UI Designer and Game Character Developer
4	Samuel Mifta samuelmifta@gmail.com	Web3 Blockchain Developer and Backend Developer
5	Eyosafet Alem eyosafetalem@gmail.com	Project Manager and Solution Architect

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